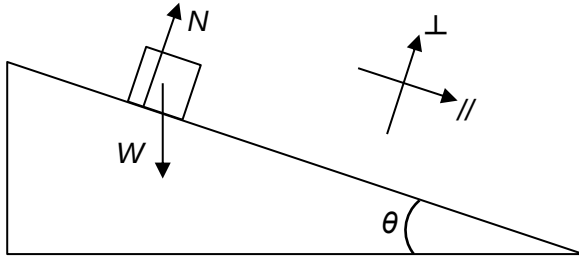


2.	Ans: D $\tan \alpha$ is unitless. Therefore, Q is unitless, and P must have the same units as $\sqrt{M}$, hence the units of P is $\text{kg}^{\frac{1}{2}}$.



3.

Ans: C



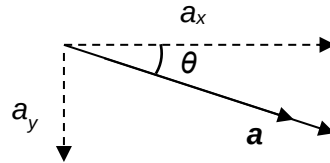
$$\vec{F}_{net} = m\vec{a}$$

$$\vec{W} + \vec{N} = m\vec{a}$$

$$\begin{pmatrix} W_{//} \\ W_{\perp} \end{pmatrix} + \begin{pmatrix} N_{//} \\ N_{\perp} \end{pmatrix} = m \begin{pmatrix} a_{//} \\ a_{\perp} \end{pmatrix}$$

$$\begin{pmatrix} mg \sin \theta \\ -mg \cos \theta \end{pmatrix} + \begin{pmatrix} 0 \\ N \end{pmatrix} = m \begin{pmatrix} a \\ 0 \end{pmatrix}$$

$$a = g \sin \theta$$



$$\begin{aligned} a_y &= a \sin \theta \\ &= (g \sin \theta)(\sin \theta) \\ &= g \sin^2 \theta \end{aligned}$$

4.

Ans: B